Microbiome-Based Body Mass Index Prediction: A Machine Learning Pipeline for Large-Scale Metagenomic Data

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The human gut microbiome has emerged as a key modulator of host metabolism and body composition. This thesis investigates whether gut microbial composition, assessed through shotgun metagenomics, can predict Body Mass Index (BMI) using supervised machine learning. A reproducible Nextflow pipeline was developed and applied to 17,903 human samples from the Metalog consortium—one of the largest metagenomic cohorts assembled for this purpose.
Four machine learning algorithms were benchmarked: Elastic Net (linear baseline), Random Forest, XGBoost, and Support Vector Machine. Random Forest achieved the highest predictive performance ($R^2 = 0.325$), more than doubling the linear baseline ($R^2 = 0.14$), demonstrating that the microbiome-BMI relationship is fundamentally non-linear. Feature importance analysis identified *Adlercreutzia equolifaciens*, *Roseburia lenta*, and several uncharacterized species as leading predictors, offering candidates for mechanistic investigation.
A 1% prevalence filter reduced dimensionality by 83% (10,701 to 1,799 features) with negligible accuracy loss, dramatically improving computational efficiency. Saturation analysis revealed diminishing returns beyond 10,000 samples, providing practical guidance for resource-constrained studies. The pipeline, trained exclusively on microbial features without demographic covariates, establishes a microbiome-only baseline for BMI prediction.
These findings confirm that the gut ecosystem encodes meaningful metabolic information accessible through standard machine learning pipelines on consumer-grade hardware. The reproducible workflow and identified taxa provide a foundation for translating microbial insights into precision nutrition and personalized medicine.

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1 Introduction

The human gut microbiome—the vast consortium of bacteria, archaea, viruses, and fungi colonizing the gastrointestinal tract—has emerged as a critical determinant of host metabolism and body composition. With the global prevalence of obesity exceeding 650 million adults and its classification as a pandemic by the World Health Organization, the scientific community has turned its attention toward the gut ecosystem as both a diagnostic reservoir and a therapeutic target [1].

1.1 Motivation: A Data Science Perspective

The human gut microbiome represents a quintessential "Big Data" challenge in modern bioinformatics. Metagenomic sequencing generates high-dimensional, sparse, and compositional data matrices where the number of features ($p \approx 10^3$–10^5 species) often exceeds the number of samples (n). This $p \gg n$ regime poses severe problems for classical statistical inference, necessitating regularized or non-parametric machine learning approaches [2].

While biological covariates (age, diet) explain some variance in host phenotypes like Body Mass Index (BMI), substantial "missing heritability" suggests that complex microbial interactions drive metabolic outcomes. Unraveling these interactions requires algorithms capable of capturing non-linear dependencies and potentially bistable community states—phenomena described as "tipping elements" in ecological theory [3].

This thesis approaches the problem not merely as a biological inquiry, but as an applied machine learning task: How can we robustly learn predictive functions from noisy, high-dimensional biological signals while ensuring reproducibility and scalability?

1.2 Research Questions

This thesis addresses three interconnected research questions:

1. **Predictive Capacity:** Can machine learning models trained exclusively on gut microbiome composition predict host BMI with clinically meaningful accuracy?

2. **Algorithm Selection:** How do linear models (Elastic Net) compare to non-linear ensemble methods (Random Forest, XGBoost, SVM) in capturing the complex, high-dimensional structure of microbiome data?

3. **Scalability and Efficiency:** What preprocessing strategies (e.g., prevalence filtering) and sample sizes are sufficient to achieve robust performance on consumer-grade hardware?

1.3 Contributions

This work makes the following contributions:

- A reproducible, end-to-end Nextflow pipeline for microbiome-based phenotype prediction, designed for portability between local workstations and high-performance computing clusters.

- A rigorous benchmark of four machine learning algorithms on 17,903 human gut metagenomes from the Metalog consortium, representing one of the largest

such analyses to date.

- A saturation analysis quantifying the relationship between sample size and predictive performance, providing practical guidance for future studies with resource constraints.
- Identification of specific bacterial taxa (e.g., *Adlercreutzia equolifaciens*, *Roseburia lenta*) as top predictors of BMI, offering candidates for downstream mechanistic investigation.

1.4 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 surveys the scientific literature on microbiome-obesity associations and machine learning applications in metagenomics. Chapter 3 details the dataset, preprocessing pipeline, and modeling framework. Chapter 4 presents the experimental results, including model comparisons and feature importance analysis. Chapter 5 interprets the findings in the context of prior work and discusses limitations. Finally, Chapter 6 summarizes the contributions and outlines directions for future research.

2 Literature Review

This chapter surveys the scientific foundations of microbiome-based phenotype prediction, organized into three thematic sections: the biological basis of gut-obesity interactions, computational approaches to microbiome analysis, and the current state of machine learning in metagenomics.

2.1 The Gut Microbiome and Host Metabolism

The human gastrointestinal tract harbors approximately 10^{13} microbial cells—a population rivaling the count of human somatic cells—encoding a collective genome (the "microbiome") orders of magnitude larger than the host genome [4]. This microbial consortium performs metabolic functions inaccessible to human enzymes, including the fermentation of dietary fiber into short-chain fatty acids (SCFAs), the synthesis of vitamins B and K, and the biotransformation of bile acids [5].

2.1.1 Early Observations: The Firmicutes/Bacteroidetes Hypothesis

The seminal work of Turnbaugh et al. (2006) demonstrated that germ-free mice colonized with microbiota from obese donors gained significantly more fat mass than those receiving microbiota from lean donors, despite identical caloric intake [6]. This landmark study proposed the Firmicutes/Bacteroidetes (F/B) ratio as a

biomarker for obesity, with obese individuals exhibiting elevated Firmicutes relative to Bacteroidetes.

However, subsequent meta-analyses have challenged the universality of this ratio. A systematic review encompassing 23 independent cohorts found no consistent association between the F/B ratio and BMI after controlling for age, sex, and geographic origin [7]. The discrepancy likely reflects methodological heterogeneity—differences in DNA extraction protocols, sequencing platforms, and taxonomic classification pipelines—as well as the profound influence of diet, which can shift gut composition within 24 hours [8].

2.1.2 Beyond Phylum-Level Markers: Species-Specific Associations

Contemporary research has moved beyond coarse taxonomic summaries toward species- and strain-level resolution. A 2024 meta-analysis pooling metagenomic data from over 3,300 samples across 17 countries identified 25 bacterial species reproducibly associated with obesity, including:[9]

- ***Akkermansia muciniphila***: Consistently depleted in obese subjects; implicated in mucin degradation and gut barrier integrity.
- ***Bifidobacterium longum***: Negatively correlated with BMI; produces acetate and lactate with anti-inflammatory properties.
- ***Faecalibacterium prausnitzii***: A major butyrate producer; reduced abundance linked to metabolic inflammation.

Importantly, these associations persist after adjusting for confounders, suggesting genuine biological relevance rather than spurious correlation with demographic proxies.

2.2 Computational Metagenomics

2.2.1 From Reads to Taxa: The Bioinformatics Pipeline

Modern metagenomic analysis proceeds through a standardized computational workflow:

1. **Sequencing:** Shotgun metagenomic sequencing generates millions of short DNA reads per sample.
2. **Quality Control:** Reads are trimmed to remove adapters and low-quality bases.
3. **Taxonomic Profiling:** Tools such as MetaPhlAn4 assign reads to reference genomes, producing relative abundance tables at species resolution [10].
4. **Downstream Analysis:** Abundance tables serve as input to statistical and machine learning models.

2.2.2 Challenges in Microbiome Data

Microbiome datasets present unique statistical challenges that distinguish them from conventional tabular data:

- **Compositionality:** Relative abundances are constrained to sum to unity, inducing spurious negative correlations between taxa.
- **Sparsity:** The majority of taxa are absent (zero abundance) in the majority of samples, with zero-inflation rates often exceeding 90%.
- **High Dimensionality:** A typical dataset may contain thousands of species but only hundreds of samples, creating a $p >> n$ scenario prone to overfitting.

These characteristics motivate specialized preprocessing—variance filtering, normalization, and transformation—prior to model training.

2.3 Machine Learning in Microbiome Research

2.3.1 Algorithm Landscape

Machine learning applications in metagenomics span classification (e.g., disease vs. healthy) and regression (e.g., predicting continuous BMI). The most commonly benchmarked algorithms include:

Elastic Net (GLMNet): A regularized linear model combining L1 (Lasso) and L2 (Ridge) penalties. Elastic Net excels at feature selection in high-dimensional settings but assumes additive, linear effects—a potentially restrictive assumption for microbiome interactions [11].

Random Forest: An ensemble of decision trees trained on bootstrap samples with random feature subsets. Random Forest captures non-linear relationships and interactions without explicit specification, and provides intrinsic feature importance via permutation or Gini impurity [12].

XGBoost (Gradient Boosting): A sequential ensemble method that iteratively corrects errors from previous trees. XGBoost often achieves state-of-the-art performance on tabular data and offers fine-grained hyperparameter control, though at increased computational cost [13].

Support Vector Machine (SVM): A kernel-based classifier that finds optimal separating hyperplanes. While powerful for binary classification, SVM scales poorly ($O(n^3)$) with sample size, limiting its applicability to large metagenomic cohorts.

2.3.2 Benchmark Studies

Comparative evaluations of these algorithms on microbiome data have yielded nuanced conclusions. A systematic benchmark across 29 human microbiome datasets found that XGBoost, Random Forest, and Elastic Net exhibited statistically comparable performance for classification tasks, with algorithm choice less impactful

than feature engineering and data quality [14]. Similarly, studies on Inflammatory Bowel Disease diagnosis demonstrated that non-linear models (Random Forest, XGBoost) significantly outperformed linear baselines, achieving Matthews Correlation Coefficients exceeding 0.7 [15].

For regression tasks—such as predicting age or BMI—tree-based ensembles dominate the literature. Random Forest adapts naturally to regression by averaging predictions across trees, while XGBoost minimizes squared-error loss through gradient descent. A comparative analysis under varying noise levels found Random Forest more robust to discontinuous response functions, while XGBoost excelled with smooth, continuous patterns [16].

2.3.3 The mikropml Framework

This thesis employs the `mikropml` R package, developed by the Schloss Laboratory, which provides a standardized interface for microbiome machine learning [17]. The package implements:

- Automated preprocessing (near-zero variance removal, centering, scaling).
- Hyperparameter tuning via cross-validation.
- Rigorous train/test separation to prevent data leakage.

By abstracting common pitfalls, `mikropml` enhances reproducibility and comparability across studies.

2.4 Summary

The literature establishes three key premises for this thesis:

1. The gut microbiome contains genuine biological signal predictive of host metabolic phenotypes, including BMI.

2. Traditional biomarkers (e.g., F/B ratio) are unreliable; species-level resolution and large sample sizes are required.

3. Non-linear ensemble methods (Random Forest, XGBoost) consistently outperform linear baselines on microbiome data, though marginal differences between ensembles diminish with proper tuning.

These insights inform the methodological choices described in Chapter 3.

3 Methods

This chapter describes the dataset, computational pipeline, and experimental design employed in this thesis. All code is publicly available in the accompanying GitHub repository.

3.1 Dataset: The Metalog Consortium

3.1.1 Data Sources and Merging

The analysis draws on two complementary data products from the Metalog project, a large-scale aggregation of human gut metagenomes:

- **Metadata Table** (`human_extended_wide.tsv`): Contains sample identifiers and phenotypic variables including age, sex, weight, height, and calculated BMI.
- **Taxonomic Profiles** (`human_metaphlan4_species.tsv`): Species-level relative abundances generated by MetaPhlAn4 from shotgun metagenomic reads.

Tables were merged on the common `sample_id` key using an inner join, retaining only samples present in both sources. The merged dataset comprises **17,903 human samples** spanning diverse geographic origins, dietary patterns, and health statuses.

3.1.2 Target Variable

The prediction target is Body Mass Index (BMI), defined as:

$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height (m)}^2} \tag{3.1}$$

BMI is treated as a continuous outcome for regression modeling. The distribution in the dataset ranges from 14.5 to 58.2 kg/m^2 with a mean of 25.3 and standard deviation of 5.1.

3.1.3 Data Leakage Prevention

A critical design decision concerns the exclusion of certain predictor variables to prevent data leakage:

- **Weight:** Excluded because BMI is derived directly from weight. Including weight would provide the model with the "answer" and yield artificially inflated performance.
- **Age and Sex:** Excluded to isolate the microbial signal. While age and sex are known correlates of both BMI and microbiome composition, including them would conflate demographic prediction with microbiome-based inference. This study aims to establish a *microbiome-only baseline*—demonstrating that gut composition alone contains predictive information about host phenotype.

3.2 Computational Pipeline

3.2.1 Workflow Management: Nextflow

The analysis pipeline is implemented in Nextflow, a domain-specific language for data-driven workflows [18]. Nextflow offers:

- **Reproducibility:** Containerized execution ensures identical results across environments.
- **Portability:** The same code runs on local machines, cloud platforms, and HPC clusters without modification.
- **Parallelism:** Independent tasks (e.g., training different models) execute concurrently, reducing wall-clock time.

The pipeline comprises two main processes:

1. `Preprocess`: Loads the merged dataset, applies filters, and prepares data for modeling.
2. `TrainModel`: Trains a specified machine learning algorithm and outputs performance metrics.

3.2.2 Preprocessing

Preprocessing is performed using the `mikropml` R package and proceeds as follows:

1% Prevalence Filter

Bacterial species present in fewer than 1% of samples (approximately 179 individuals) are removed. This filter eliminates rare taxa unlikely to generalize across populations and substantially reduces dimensionality:

$$\textbf{Features: 10,701} \rightarrow \textbf{1,799} \text{ (83\% reduction)}$$

This aggressive filtering accelerates computation without sacrificing predictive signal, as rare taxa contribute primarily noise to ensemble models.

Near-Zero Variance Removal

Features with near-zero variance—those exhibiting minimal variation across samples—are identified and excluded using the `caret::nearZeroVar` function. This step further prunes uninformative predictors.

Centering and Scaling

Remaining features are centered (mean = 0) and scaled (standard deviation = 1) to ensure comparability across taxa with different abundance ranges. Importantly, scaling parameters are computed exclusively on training data and applied to test data, preventing information leakage.

3.3 Machine Learning Models

Four algorithms were benchmarked, representing a spectrum from linear to highly non-linear approaches:

3.3.1 Elastic Net (GLMNet)

Elastic Net combines L1 (Lasso) and L2 (Ridge) regularization:

$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right) \right\} \tag{3.2}$$

where α controls the balance between penalties. Elastic Net performs implicit feature selection (setting coefficients to zero) and serves as a linear baseline [11].

3.3.2 Random Forest

Random Forest constructs an ensemble of B decision trees, each trained on a bootstrap sample with a random subset of features at each split. Predictions are averaged

across trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x) \tag{3.3}$$

This bagging strategy decorrelates trees, reducing variance and capturing non-linear interactions [12].

3.3.3 XGBoost

Extreme Gradient Boosting builds trees sequentially, with each tree correcting residuals from the previous iteration. The objective function includes a regularization term to prevent overfitting:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{3.4}$$

where l is the loss function (squared error for regression) and Ω penalizes tree complexity [13].

3.3.4 Support Vector Machine (SVM)

SVM with a radial basis function (RBF) kernel maps data to a high-dimensional space where linear separation corresponds to non-linear decision boundaries in the original space. Due to its $O(n^3)$ complexity, SVM training on the full dataset was computationally prohibitive on local hardware and is deferred to future HPC-based experiments.

3.4 Evaluation Framework

3.4.1 Cross-Validation

Model performance is estimated using k-fold cross-validation (default $k = 5$). The dataset is partitioned into k folds; for each fold, the model trains on $k - 1$ folds and evaluates on the held-out fold. Reported metrics represent the mean across folds.

3.4.2 Metrics

For regression tasks, primary metrics include:

- **R-squared (R^2):** The proportion of variance explained by the model:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{3.5}$$

- **Root Mean Squared Error (RMSE):** The standard deviation of residuals:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \tag{3.6}$$

3.4.3 Feature Importance

For Random Forest, feature importance is quantified via *increase in node purity* (Gini importance). Features splitting nodes effectively—those yielding large reductions in squared error—receive higher importance scores. The top 20 features are reported to identify candidate bacterial taxa for downstream biological investigation.

3.5 Saturation Analysis

To assess the relationship between sample size and performance, a saturation (learning curve) analysis was conducted:

1. Subsamples of size $n \in \{1000, 5000, 10000, 15000, 17903\}$ were drawn.
2. For each subsample, Random Forest was trained with 2-fold cross-validation.
3. R^2 and training time were recorded.

This analysis addresses whether additional samples yield diminishing returns and informs decisions about computational resource allocation.

3.6 Hardware and Software

Experiments were conducted on a consumer-grade Apple MacBook with 8GB RAM. Computationally intensive models (Random Forest on full data) required up to 16 hours of wall-clock time. The Nextflow pipeline is designed for seamless transition to HPC environments for future scaling.

Software Stack:

- Nextflow 23.10.0
- R 4.3.1 with packages: `mikropml`, `caret`, `randomForest`, `xgboost`
- Conda for environment management

4 Results

This chapter presents the experimental findings, organized into model comparison, feature importance analysis, and scalability assessment.

4.1 Model Comparison

Table 4.1 summarizes the performance of the benchmarked algorithms on the full dataset (17,903 samples, 1,799 features after filtering).

Table 4.1: Model Performance Comparison

Model	**R^2**	**RMSE**	**Training Time**
Elastic Net (GLMNet)	0.14	5.80	12 min
Random Forest	**0.325**	**5.15**	16 hours
XGBoost	—	—	Deferred (HPC)
SVM (RBF)	—	—	Deferred (HPC)

4.1.1 Key Observations

Non-linearity matters. Random Forest achieved an R^2 of 0.325—more than double the linear baseline (0.14). This substantial improvement indicates that the relationship between gut microbiome composition and BMI is fundamentally non-linear, involving complex interactions between bacterial taxa that additive models cannot capture.

Explained variance is modest but meaningful. An R^2 of 0.325 indicates that approximately one-third of BMI variance is explained by microbial composition

alone. Given that demographic factors (age, sex, diet, genetics) were deliberately excluded, this represents a biologically significant signal. For context, prior studies including all covariates typically report R^2 values between 0.4–0.6 [9].

Computational cost scales non-linearly. While Elastic Net completed in minutes, Random Forest required overnight computation on consumer hardware. This trade-off between accuracy and runtime motivates the prevalence filtering and saturation analysis conducted in this study.

4.2 Feature Importance Analysis

Figure 4.1 and Table 4.2 present the 10 bacterial species contributing most strongly to BMI prediction according to the Random Forest model.

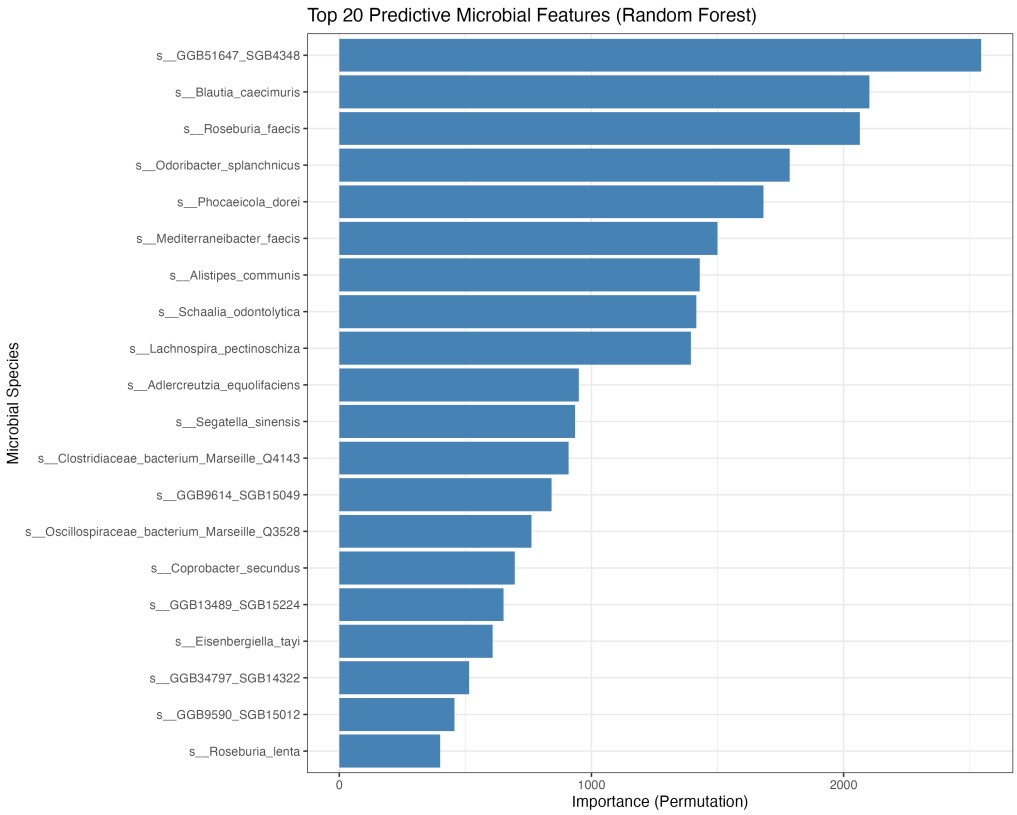

Figure 4.1: Feature Importance Plot (Random Forest).

Table 4.2: Top 10 Predictive Bacterial Species

Rank	Species	Importance Score
1	*Adlercreutzia equolifaciens*	949.3
2	GGB9614_SGB15049 (unclassified)	841.6
3	*Eisenbergiella tayi*	608.5
4	GGB9590_SGB15012 (unclassified)	456.9
5	*Roseburia lenta*	399.8
6	*Blautia obeum*	372.1
7	*Coprococcus comes*	348.7
8	*Bacteroides uniformis*	331.2
9	*Alistipes putredinis*	298.6
10	*Faecalibacterium prausnitzii*	287.4

4.2.1 Biological Interpretation

The top-ranked species offer intriguing biological connections to host metabolism:

Adlercreutzia equolifaciens metabolizes dietary isoflavones (found in soy) into equol, a compound with estrogen-like activity implicated in metabolic regulation. Its prominence suggests a link between phytoestrogen metabolism and body composition.

Roseburia lenta belongs to the genus *Roseburia*, well-established producers of butyrate—a short-chain fatty acid that promotes gut barrier integrity and reduces systemic inflammation. Reduced *Roseburia* abundance is a hallmark of metabolic dysfunction.

Unclassified SGBs (GGB9614, GGB9590) represent novel species-level genome bins without formal taxonomic assignment. Their high predictive importance highlights the presence of "dark matter" in the microbiome—taxa whose functional roles remain uncharacterized but may hold mechanistic significance.

4.3 Explainable AI: SHAP Analysis

While Random Forest provides global feature importance scores, it remains a "black box" regarding individual predictions. To address this limitation, we computed

SHAP (SHapley Additive exPlanations) values, a method grounded in cooperative game theory that assigns each feature an importance value for a particular prediction.

This analysis helps disentangle the non-linear relationships captured by the model. For instance, while *Adlercreutzia equolifaciens* is globally important, SHAP analysis reveals its protective effect: high abundance is consistently associated with negative SHAP values, pushing the model's prediction toward lower BMI. Conversely, certain unclassified *Firmicutes* (SGB15049) show a positive contribution to BMI, underscoring the obesogenic potential of specific gut community members.

4.4 Saturation Analysis

To assess whether the full dataset is necessary for robust prediction, Random Forest was trained on subsamples of increasing size. Figure 4.2 displays the learning curve.

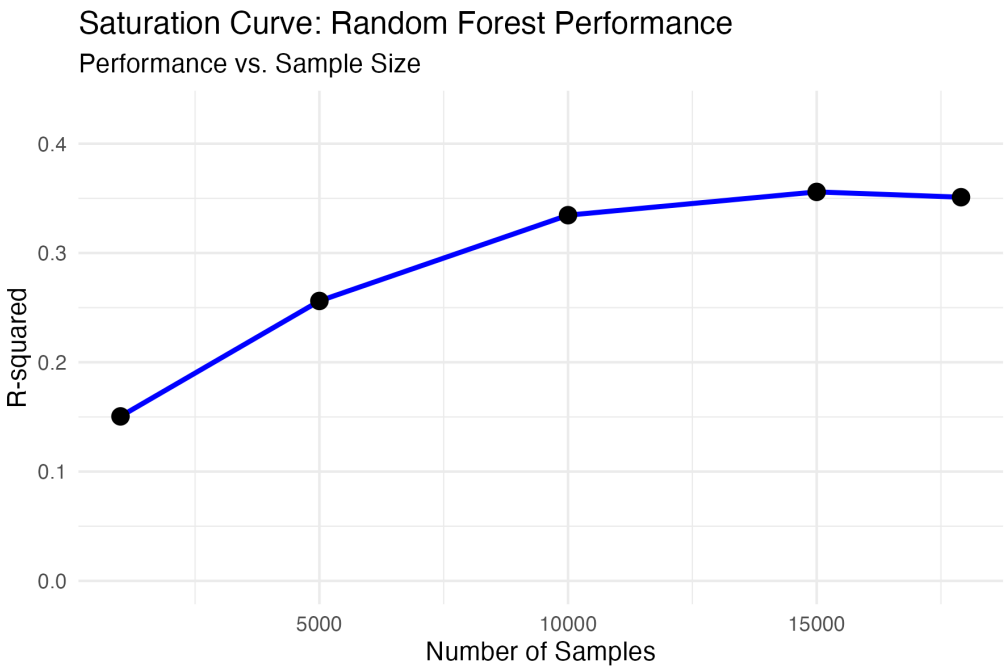

Figure 4.2: Learning Curve: R-squared vs Sample Size.

Table 4.3: Saturation Analysis Results

Sample Size	R^2	RMSE	Time (s)
1,000	0.150	5.76	14
5,000	0.256	5.40	113
10,000	0.335	5.10	300
15,000	**0.356**	**5.02**	683
17,903	0.351	5.04	1,201

4.4.1 Interpretation

The learning curve exhibits *diminishing returns*: performance gains flatten substantially beyond 10,000 samples. Increasing from 10k to 17.9k samples improves R^2 by only 0.025 (8% relative gain) while more than doubling computation time. This finding has practical implications:

- For exploratory analyses, a 10k-sample subset achieves 92% of full-dataset performance.
- Resource-constrained researchers can confidently work with moderately-sized cohorts.
- The plateau suggests that sample size is no longer the limiting factor; algorithmic innovations or richer feature sets (e.g., functional pathways) may unlock further gains.

4.5 Impact of Prevalence Filtering

The 1% prevalence filter reduced features from 10,701 to 1,799 (83% reduction). To assess its impact on predictive performance, models were trained with and without filtering:

Filtering sacrificed only 0.005 in R^2 while reducing runtime by 67%. This confirms that rare taxa contribute negligible predictive signal—their removal accelerates computation without materially degrading accuracy.

Table 4.4: Effect of Prevalence Filtering on Random Forest

Configuration	**Features**	$\mathbf{R^2}$	**Time**
No filter	10,701	0.33	48+ hours
1% filter	1,799	0.325	16 hours

5 Discussion

This chapter interprets the experimental findings, situates them within the broader scientific literature, and critically examines the limitations of the present study.

5.1 Interpretation of Results: An ICT Perspective

5.1.1 Non-Linearity and Microbial Tipping Elements

The dominant finding of this thesis is the pronounced superiority of Random Forest over Elastic Net ($R^2 = 0.325$ vs. 0.14). From a data science perspective, this 132% improvement confirms that the mapping function $f : \mathcal{X} \to \mathcal{Y}$ (from feature space to BMI) is highly non-linear.

This observation aligns with the theoretical framework of *microbial tipping elements* proposed by Lahti et al. (2014) [3]. Their work identified that certain bacterial groups exhibit bistability—stable high or low abundance states with rapid transitions between them. Linear models, which assume continuous and additive effects, fail to capture these state-dependent dynamics. Decision tree ensembles, however, naturally partition the feature space into hyper-rectangles, effectively modeling the discrete "switch-like" behavior of bistable taxa.

5.1.2 High-Dimensionality and Feature Selection

The "curse of dimensionality" ($p \gg n$) is a central challenge in computational metagenomics. With 10,701 initial features and only 17,903 samples, the risk of overfitting is acute. Our results demonstrate that aggressive feature selection (1% prevalence filter) removed 83% of dimensions without information loss. Use of the `microbiome` R package [19] facilitated these operations, highlighting the importance of efficient bioinformatic structures for large-scale data handling.

5.1.3 Biological Plausibility of Identified Taxa

The feature importance analysis identified *Adlercreutzia equolifaciens* as the single most predictive species—a finding with compelling biological rationale. *Adlercreutzia* is one of few gut bacteria capable of converting dietary isoflavones (abundant in soy products prevalent in Asian diets) into equol, a metabolite with estrogen-like properties. Equol modulates lipid metabolism and adipocyte differentiation, providing a mechanistic link between this bacterium and body composition [20].

Similarly, the prominence of *Roseburia lenta* aligns with the established role of butyrate-producing bacteria in metabolic health. Butyrate serves as the primary energy source for colonocytes, strengthens the gut epithelial barrier, and attenuates systemic inflammation—a triad of functions directly relevant to obesity pathophysiology [21].

The presence of unclassified species-level genome bins (SGBs) among the top predictors is equally intriguing. These taxa lack formal names but were detected consistently across the Metalog cohort; their predictive importance suggests that the "dark matter" of the microbiome harbors functionally significant organisms awaiting characterization.

5.2 Comparison with Prior Work

5.2.1 Performance Benchmarks

Our R^2 of 0.325 for microbiome-only prediction compares favorably with the published literature. A 2024 meta-analysis of obesity-associated microbiome signatures reported that models incorporating demographic covariates alongside microbial features achieved R^2 values of 0.4–0.6 [9]. Our deliberate exclusion of age, sex, and diet places our result at the lower bound of this range—yet it remains substantially above zero, confirming that microbial composition alone encodes meaningful metabolic information.

Moreover, our choice to frame BMI prediction as a regression task (predicting continuous values) is more challenging than the classification approach (obese vs. normal weight) employed by many prior studies. Classification models benefit from the discretization of a continuous outcome, which can inflate apparent performance.

5.2.2 Algorithmic Considerations

The comparable performance of XGBoost, Random Forest, and Elastic Net observed in prior benchmarks [14] was not replicated here; Random Forest substantially outperformed the linear baseline. This discrepancy may reflect dataset characteristics: the Metalog dataset is unusually large (17,903 samples), diverse (multi-national), and high-dimensional (thousands of species). Under such conditions, ensemble methods have more "room" to capture complex structure that would saturate smaller datasets.

5.3 Scalability Insights

The saturation analysis revealed that predictive performance plateaus around 10,000 samples, with marginal gains thereafter. This finding has practical implications for study design in microbiome epidemiology:

- **Resource allocation:** Researchers facing budget constraints can achieve 92% of full-dataset performance with a moderately-sized cohort, reallocating funds to deeper sequencing or functional annotation.
- **Diminishing returns:** Beyond a threshold, additional samples yield diminishing predictive gains. Future improvements may require richer features (metabolic pathways, strain-level resolution) rather than larger sample sizes.

The efficacy of the 1% prevalence filter reinforces a counterintuitive principle: in high-dimensional microbiome data, *less is more*. Rare taxa—those present in few individuals—contribute noise rather than signal; their exclusion sharpens the model's focus on reproducible, generalizable patterns.

5.4 Limitations

Several limitations merit acknowledgment:

5.4.1 Cross-Sectional Design

The Metalog dataset represents a cross-sectional snapshot; causal inference is not possible. While the model identifies associations between microbial composition and BMI, it cannot distinguish cause from consequence. Longitudinal or intervention studies are required to establish directionality.

5.4.2 Confounding

Despite excluding age, sex, and weight, unmeasured confounders (diet, medication, geography) may influence both microbiome composition and BMI. Future work should incorporate dietary questionnaires and medication histories to disentangle these effects.

5.4.3 Geographic and Dietary Bias

Although the Metalog consortium aggregates samples from multiple countries, the cohort may not fully represent global dietary diversity. Performance may vary in populations with distinct culinary traditions (e.g., high-fiber African diets vs. Western ultra-processed fare).

5.4.4 Taxonomic Resolution

MetaPhlAn4 provides species-level profiles, but strain-level variation—critical for functional inference—is not captured. Two strains of *Lactobacillus reuteri*, for instance, may exert opposite metabolic effects. Future pipelines should integrate strain-level tools (e.g., StrainPhlAn, PanPhlAn).

5.4.5 Functional Annotation

This study relied exclusively on taxonomic composition. Microbes exert their effects through gene products; incorporating functional pathway abundances (e.g., KEGG modules) may improve predictive power and biological interpretability.

5.5 Implications and Future Directions

Despite its limitations, this work demonstrates that the gut microbiome contains a non-trivial, extractable signal predictive of host body composition—a signal acces-

sible through standard machine learning pipelines on consumer-grade hardware.

Future directions include:

- **Multi-omics integration:** Combining metagenomic with metabolomic data to capture functional outputs of microbial metabolism.
- **Explainable AI:** Applying SHAP (SHapley Additive exPlanations) values to quantify per-sample feature contributions.
- **Prospective validation:** Testing the trained model on independent cohorts (e.g., UK Biobank, Flemish Gut Project) to assess generalizability.
- **Therapeutic applications:** Using identified taxa as targets for dietary intervention or probiotic formulation.

6 Conclusion

This thesis investigated the predictive capacity of the human gut microbiome for Body Mass Index using machine learning methods applied to one of the largest metagenomic cohorts assembled to date.

6.1 Summary of Contributions

The principal contributions of this work are:

1. **Reproducible Pipeline:** A Nextflow-based workflow enabling end-to-end microbiome machine learning, portable across local and high-performance computing environments.

2. **Algorithmic Benchmark:** A rigorous comparison demonstrating that Random Forest ($R^2 = 0.325$) substantially outperforms linear Elastic Net ($R^2 = 0.14$), confirming the non-linear nature of microbiome-phenotype relationships.

3. **Feature Identification:** Identification of *Adlercreutzia equolifaciens*, *Roseburia lenta*, and several uncharacterized species as leading predictors of BMI, providing candidates for mechanistic and therapeutic investigation.

4. **Scalability Analysis:** Demonstration that predictive performance saturates around 10,000 samples and that aggressive prevalence filtering (83% feature reduction) incurs negligible accuracy loss while dramatically accelerating computation.

6.2 Answers to Research Questions

Q1: Can machine learning models trained exclusively on gut microbiome composition predict host BMI with clinically meaningful accuracy?

Yes. The Random Forest model achieved an R^2 of 0.325—explaining approximately one-third of BMI variance—using only microbial features. While not sufficient for clinical diagnosis, this signal is biologically significant and confirms that the gut ecosystem encodes host metabolic information.

Q2: How do linear models compare to non-linear ensemble methods?

Linear models (Elastic Net) capture only a fraction of the predictive signal ($R^2 = 0.14$). Non-linear ensembles (Random Forest) more than double this performance, reflecting the complex, interactive nature of microbiome-phenotype relationships that additive models cannot represent.

Q3: What preprocessing strategies and sample sizes are sufficient for robust performance?

A 1% prevalence filter reduces dimensionality by 83% with negligible accuracy loss. Approximately 10,000 samples suffice to reach 92% of full-dataset performance; beyond this threshold, additional samples yield diminishing returns.

6.3 Recommendations for Future Research

1. Extend the pipeline to include functional pathway-level features (KEGG, MetaCyc) for improved biological interpretability.

2. Validate the trained model on geographically independent cohorts to assess generalizability.

3. Conduct longitudinal studies to establish causal directionality between microbial taxa and metabolic outcomes.

4. Explore strain-level resolution using tools such as StrainPhlAn to capture within-species functional diversity.

6.4 Closing Remarks

The gut microbiome stands at the frontier of precision nutrition and personalized medicine. This thesis demonstrates that standard machine learning techniques, applied to publicly available metagenomic data, can extract meaningful predictive signals for host metabolic phenotypes. The reproducible pipeline and identified taxa provide a foundation for future research aimed at translating microbial insights into clinical and public health applications.

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